

AJAY BARMAN

Full-Stack Developer | AI Agent & Backend Engineer

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SUMMARY

Full-Stack Developer and Computer Science undergraduate specializing in Next.js, TypeScript, Node.js, PostgreSQL, and MongoDB. Built and deployed production-grade AI-powered, real-time, and backend systems across web applications, cloud infrastructure, authentication, and CI/CD.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Rust, Python

Technologies & Frameworks: React.js, Next.js, Node.js, Express.js, TanStack Query, Socket.io, WebRTC, Tailwind CSS, Prisma, Drizzle, Mongoose

Databases & Cloud: PostgreSQL, MongoDB, AWS (EC2), Nginx, Docker, Vercel, Neon, Cloudinary

AI & Developer Tools: AgentKit, Inngest, OpenRouter, Neovim, Postman, Cursor, Git, GitHub

EXPERIENCE

Independent Software Engineer | Self-Directed Development

Feb 2024 – Present

- Built and deployed **10+ production-grade full-stack applications** spanning **AI app generation, online code execution, and real-time communication** using Next.js, React.js, TypeScript, Node.js, PostgreSQL, MongoDB, AWS, Docker, and Nginx.
- Delivered systems generating **100+ applications**, executing code across **6 languages**, and supporting **100+ REST endpoints** and **22 real-time socket events**, owning architecture, backend, infrastructure, security, and deployment end-to-end.

PROJECTS

V0 Clone | Next.js, TanStack Query, Inngest, E2B, AgentKit, OpenAI API, Clerk(OAuth), Prisma, PostgreSQL(Neon)

Aug 2026 – Sep 2026

- Built and shipped an **agentic AI app-builder** that turns plain-language website descriptions into production-ready web apps, generating **100+ apps** for multiple users with a **78% build-success rate** and **~4 min** median time-to-first-version.
- Engineered a **multi-agent build pipeline** on Inngest + E2B sandboxes, orchestrating a **4-tool code-agent network** (terminal, file read/write, build verification) across up to **10 retry iterations** gated by an automated **tsc** type-check, cutting failed builds by **~40%**.
- Optimized LLM spend with **OpenAI prompt caching** plus an “edit prior build” memory model that re-injects prior generated files into fresh sandboxes, reducing per-rebuild token cost **~55%** at a **~70% cached-input hit rate**.
- Shipped production infra end-to-end: **Clerk OAuth**, a credit/rate-limit system that throttled LLM calls and curbed abusive API spend by **~30%**, GitOps-style CI/CD to Vercel with Inngest Cloud sync, and Neon-hosted Postgres at **99.9% uptime** / **~50ms p50 query latency**.

CodeLeet | Next.js, Clerk(OAuth), Judge0, MonacoEditor, AWS, Nginx, Prisma, PostgreSQL (Neon)

May 2026 – Aug 2026

- Architected CodeLeet, a full-stack **Next.js 16 + TypeScript** LeetCode clone, with a self-hosted **Judge0 engine on AWS EC2** executing code in **6 languages** (Python, Java, C++, JavaScript, TypeScript, Rust) across 15 problems (7 Easy / 6 Medium / 2 Hard), batch-running up to 6 hidden test cases per submission with **sub-200ms execution** (avg **0.15s**, p95 **0.16s**), replacing paid Judge0 API entirely.
- Built a **Monaco-powered IDE** with tabbed testcase/result panels (auto-switch, animated run states), per-user progress tracking across 15 problems, expandable submission history, and playlist/bookmark CRUD gated by server-side **Clerk sessions and RBAC** (admin-only problem creation enforced in API routes), persisted across **7 relational models** in Postgres.
- Hardened and deployed a **multi-service stack**: Next.js under PM2 behind Nginx (port 80 only), Dockerized Judge0, PostgreSQL, and Redis, Clerk sessions, and Hostinger DNS with the entire stack (app + sandbox + DB + workers) running on a single **1 vCPU / 1 GB RAM t4g.micro** with 30 GB EBS, live in production for under **~\$10/mo**.

Threadly | React.js, Turborepo, Express.js, MongoDB, Mongoose, Cloudinary, Socket.io, WebRTC, Mediasoup

Feb 2025 – Jul 2026

- Engineered a **real-time chat platform** (“Threadly”) with Socket.io, WebRTC, and a **mediasoup SFU**, delivering instant messaging, typing indicators, and 1:1 & group video/audio calls across **35 REST endpoints** and **22 real-time socket events**, supporting **3+ concurrent call participants** with low-latency P2P relay and a collaborative canvas whiteboard with live stroke sync and persistence.
- Implemented **hardened authentication** with JWT access/refresh token rotation (24h/10-day expiry), bcrypt hashing, and httpOnly cookies; defended file handling against **SSRF attacks** via backend-proxied Cloudinary downloads and applied rate limiting (**5,000 requests/15 min per IP**) to block abuse and brute-force attempts, reducing security-critical attack surface across the full upload/download pipeline.
- Architected a **pnpm + Turborepo monorepo** (2 apps, 3 shared packages, **~11,500 LOC**) with strict TypeScript across the board, reusable ESLint/tsconfig presets, and a unified dev/build/lint/type-check pipeline, cutting per-feature setup overhead and enabling isolated package builds via Turborepo’s task cache.

ACHIEVEMENTS & CERTIFICATIONS

- Problem Solving:** Solved 300+ problems across LeetCode and NeetCode by applying data structures, algorithm design, and complexity analysis to optimize runtime and memory performance.
- Certificates:** LLM Engineering, RAG, & AI Agents - Udemy; Complete React and NextJS course with AI-powered projects - Udemy.
- Leadership:** Spearheaded a cross-functional team through two consecutive Smart India Hackathons, securing a spot in the **top 15%** (out of 200+ teams) of competitors in the Inter-College rounds.

EDUCATION

ABES INSTITUTE OF TECHNOLOGY

Ghaziabad, India

Bachelor of Technology, Major in Computer Science

Jul 2023 – Jul 2027

Cumulative GPA: 7.5/10.0(Recent Semester GPA: 8.14/10.0)

JKG INTERNATIONAL SCHOOL VIJAYANAGAR

Ghaziabad, India

CBSE XII PCM

Apr 2020 – Apr 2022

Score: 86.88%